

Math for ML

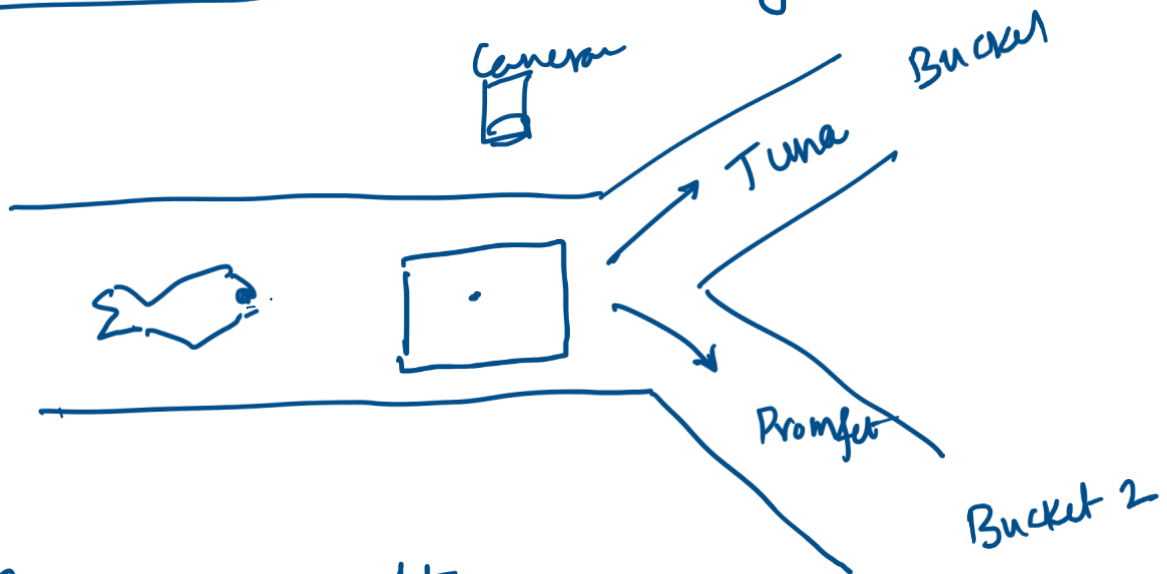
10 Lectures — 5 - linear Algebra
coord geometry
5 - Calculus

Introduction

ML

Fish Classifier

Fish Sorting machine



Size		weight	Type
length	width	weight	
20	10 cm	1.2 kg	Pomfret
20	15	2.5 kg	Tuna

Input variable features

Terminology

Data / Dataset

Say you will learn AI, nobody panics



But say you will learn linear algebra and optimisation, then everybody loses their minds

- Linear Algebra
- Coord Geometry
- Calculus

→ Probability

→ Statistics

x_1 x_2 x_3 x_4

features

Input variables

Independent Variables

Predictors

x_1	x_2	x_3	x_4	yType
x	x	x	x	Tuna
				Pomfret

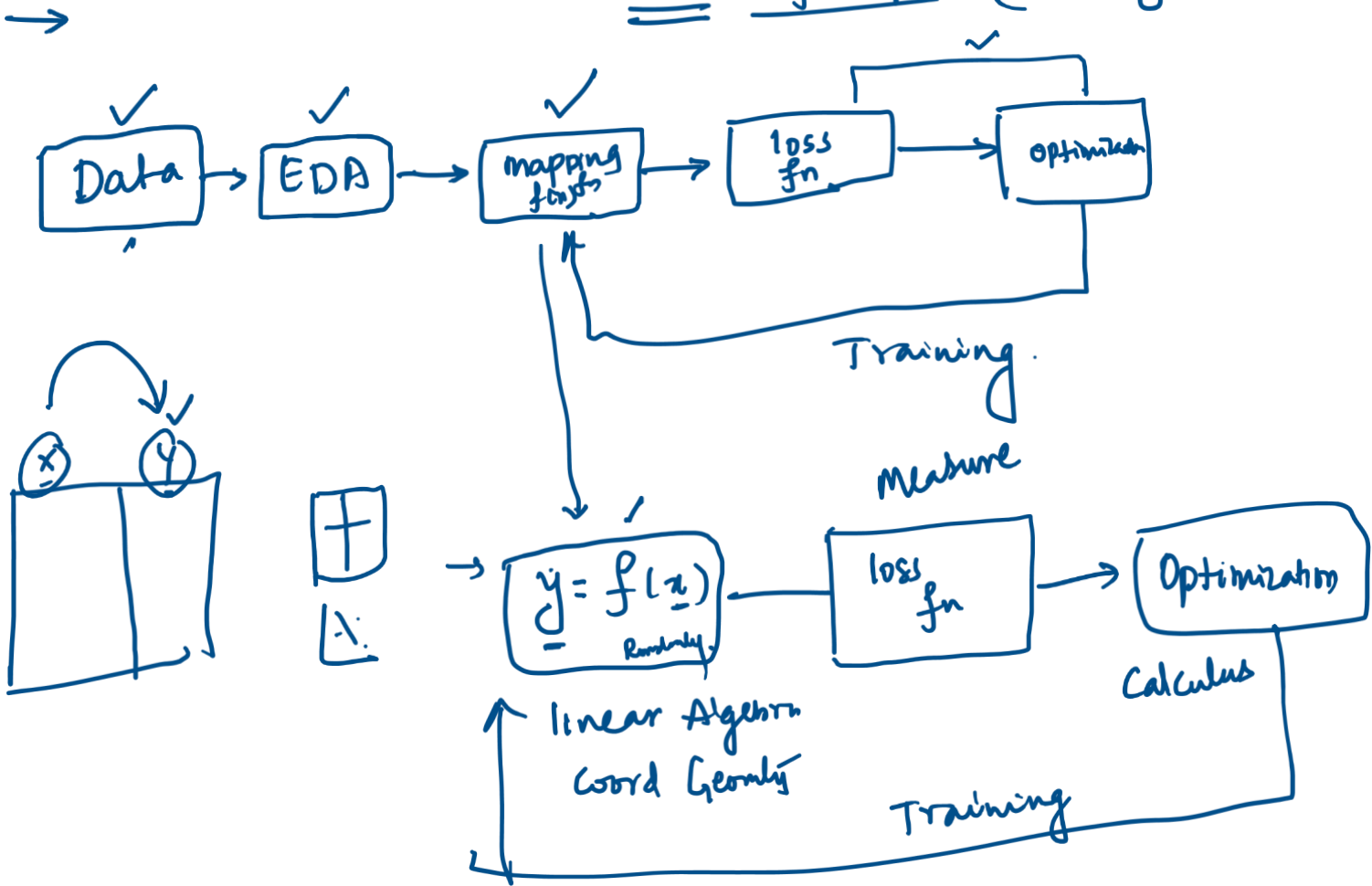
labelled
Database

Data point / observation

Target variable
label
Dependent variable

Predicted variable

ML life cycle (Training)



$$y = mx + c$$

m?, c?

$$\hat{y} = m(x) + c$$

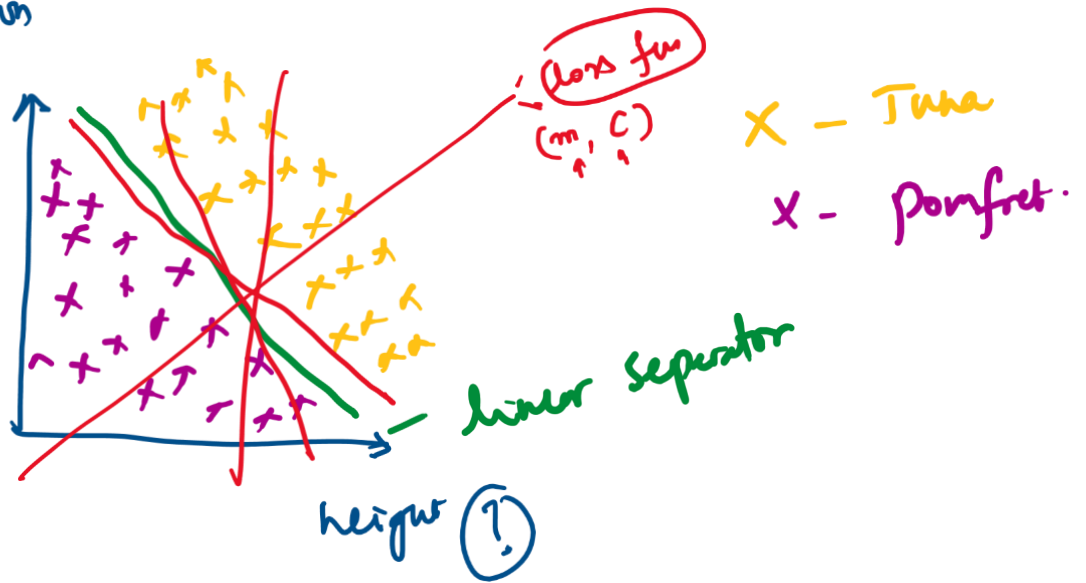
→ Classification (Binary Classification)

height	weight	Gender	T-shirt size	Target
6	80	M	L	L
5.8	70	M	M	M
			S	S
			XL	XL

Input

Multiclass classification

Binary Classification
width



length	width	weight	Target Type
			Tuna

Target $\rightarrow$ dependent on 'x'

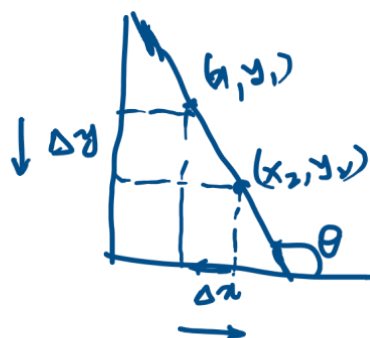
X - height, weight, width
independent variables

Mapping function

$$y = mx + c$$

$$m = \tan \theta = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{\Delta y}{\Delta x}$$



Slope $m = \frac{y_2 - y_1}{x_2 - x_1}$

Slope ∞  $\frac{\Delta y}{\Delta x = 0}$

$y = mx + c$? m ?

Lg: $w_1 x_1 + w_2 x_2 + w_0 = 0$ ✓

→ $y = -\frac{w_0}{w_2} - \frac{w_1}{w_2} x_1$ $y = mx + c$

$m \rightarrow -\frac{w_1}{w_2}$ $c = -\frac{w_0}{w_2}$

$w \rightarrow$ weights

$y = mx + c$ ✓

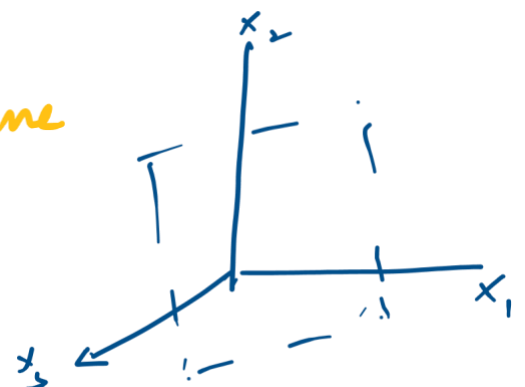
Lg: $w_0 + w_1 x_1 + w_2 x_2 \rightarrow$ 2D plane

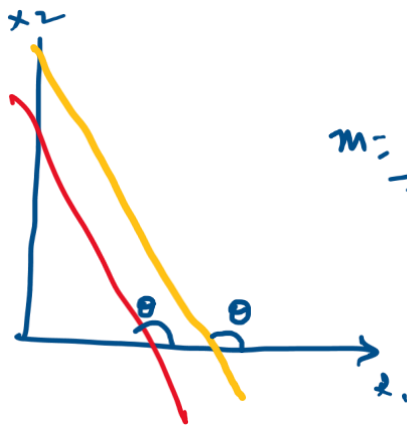
$w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 \rightarrow$ 3D plane

→ $w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 \rightarrow$ 4D - Hyperplane



2D - plane

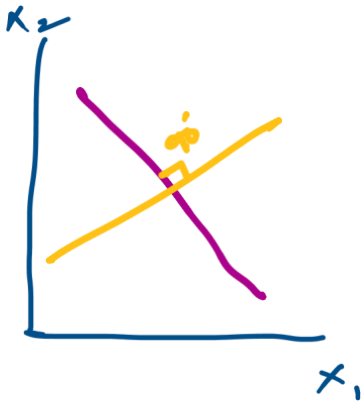




$$m = \tan \theta$$

Planes - Parallel.

$$m_1 = m_2$$



Planes - Perpendicular

$$m_1 \times m_2 = -1$$

Classification (ML task)

$$y = mx + c$$



✓ 2D-plane?

$$w_0 + w_1 x_1 + w_2 x_2 = 0$$

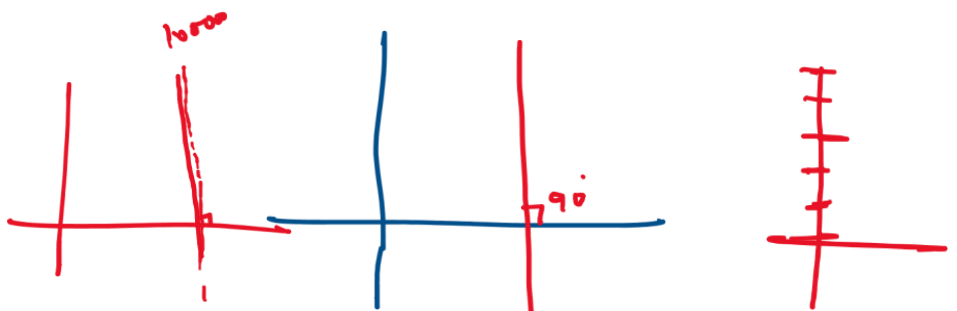
$$3D \quad w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 = 0$$

$$nd \quad w_0 + w_1 x_1 + \dots + w_n x_n = 0$$

nd

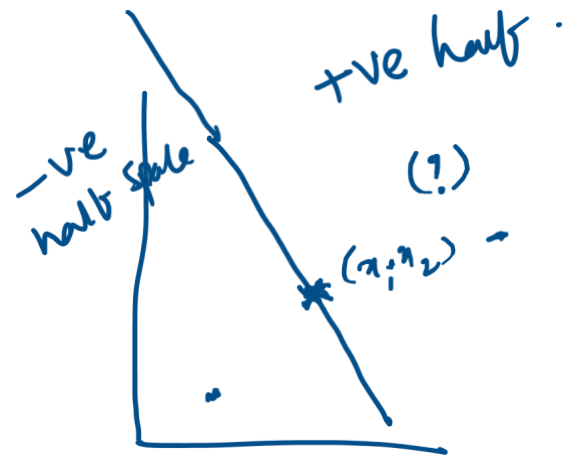
$$\rightarrow y = mx + c$$

$$m \rightarrow \infty$$





x - Tuna
x - Pomfret



Half Spaces

$$w_0 + w_1 x_1 + w_2 x_2 = 0$$

$$w_0 + w_1 x_1 + w_2 x_2 > 0$$

$$w_0 + w_1 x_1 + w_2 x_2 < 0$$

high width
(?) (25), (15) cm

$$y = f(x) = w_0 + w_1 x_1 + w_2 x_2$$

✓ $f(x) > 0 \rightarrow$ Tuna
✓ $f(x) < 0$ Pomfret
 $f(x) = 0$ - ?

weights $\Rightarrow$ $w_0 \quad w_1 \quad w_2$

$y = \text{Tuna}$ if $f(x) \geq 0$
 $y = \text{Pomfret}$ if $f(x) < 0$

